

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (EC) Sem - IV

University Examination, 2010-11

EC - 207 Control Systems

Date: 11.05.11, Wednesday
Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

- (1). All questions are compulsory.
- (2). Indicate clearly, the option you attempt along with its respective question number.
- (3). Figures to right indicate marks
- (4). Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION I

Q:1

- a) Discuss the differences between open loop control system and close loop control system. [02]
- b) Write short note on comparison between root locus technique and nyquist stability criteria for stability study of control systems. [03]
- c) Determine $\frac{C_1(s)}{R_1(s)}$, $\frac{C_2(s)}{R_2(s)}$ for multiple input system shown in figure(a). [06]

Q:2

- a) Draw the root locus of the control system whose transfer function is following as K varies 0 to ∞ [07]

$$G(s)H(s) = \frac{K}{s(s^2 + 2s + 2)}$$

- b) Explain the derivation of peak time, maximum peak overshoot and rise time. [05]

$\frac{\pi}{\omega_d}$ OR $\frac{-\pi}{\omega_d}$ $\frac{\pi - \theta}{\omega_d}$

Q:2

- a) Draw the root locus of the control system whose transfer function is following as K varies 0 to ∞ [07]

$$G(s)H(s) = \frac{K(s+4)}{s(s^2+2s+2)}$$

- b) For a system $G(s)H(s) = \frac{K}{s^2(s+2)(s+3)}$, find the value of K to limit [05]

steady state error to 10 when input to system is $1+10t+\frac{40}{2}t^2$

Q:3

- a) Determine the stability of the closed loop unity feedback system using [06]
Nyquist stability criterion for system whose open loop transfer function

$$G(s) = \frac{1}{s^2(1+s)(1+2s)}$$

- b) Find $\frac{C(s)}{R(s)}$ of the figure (b) using signal flow graph. [06]

Q:3

OR

- a) Discuss the stability of the system using nyquist plot for [06]

$$G(s)H(s) = \frac{20}{s(s+4)(s-2)}$$

- b) Using mason's gain formula, find $\frac{X_2}{X_1}$ of the figure (c). [06]

$$T(s) = \frac{C \text{ Adj } \{sI - A\}_{13}}{\det(sI - A)} + D =$$

SECTION II

Q:4

- a) Derive the expression of transfer function of the system which is [05]
represented in the following standard space form: [14.7]

$$\dot{x} = Ax + Bu, \quad y = Cx + Du \quad L.T$$

- b) Derive the transfer function for field control DC servo motor. [06]

Q:5

- a) For the mechanical system shown in figure (d), draw the force-voltage [08]
and force-current electrical analogous circuits.

- b) Determine the range of values of K for which the unity feedback system [04]
whose open loop

$$T.F = \frac{K}{s(s^2+s+1)(s+4)}$$

is stable and determine the frequency of

$$0 < K < 3.36$$

stable.

sustained oscillations.

OR

- ✓ a) For the mechanical system shown in figure (e), draw the torque-voltage [08]
and torque-current electrical analogous circuit.
- ✓ b) The characteristic equation of a system is given by [04]
 $S^4 + 8S^3 + 18S^2 + 16S + 4 = 0$. Determine the stability of a system by means
of Hurwitz criterion

Q:6

717
Ex-10

- a) A system has $G(s)H(s) = \frac{K(s+2)}{s(s+4)(s+10)}$. Using bode plot find K to [08]
get P.M. = +30° $K = 563.67$ $K' = 28.183$
- b) For the electrical network shown in figure (f), determine the state [04]
equations.

OR

Q:6

710
Ex-8

- a) A unity feedback system has $G(s) = \frac{3(s+1)(s+700)}{s^2(s^2 + 18s + 400)}$. Draw the bode [08]
plot and comment on stability. $K = 5.25$
- b) Explain the following terms: [04]
- Relative stability
 - Zero input stability

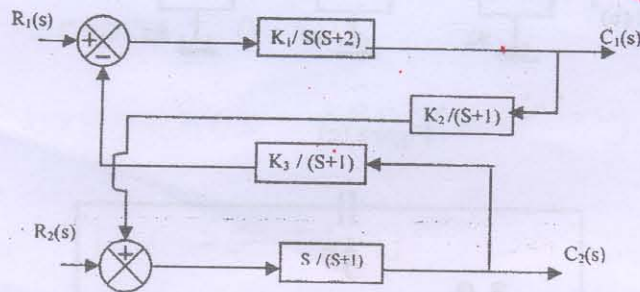


Figure (a)

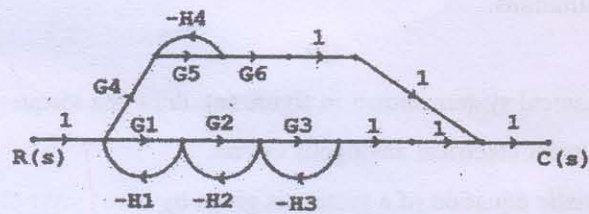


Figure (b)

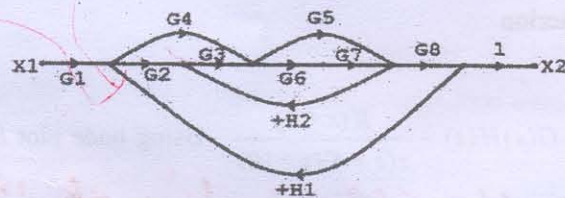


Figure (c)

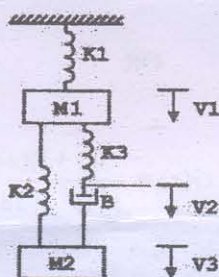


Figure (d)

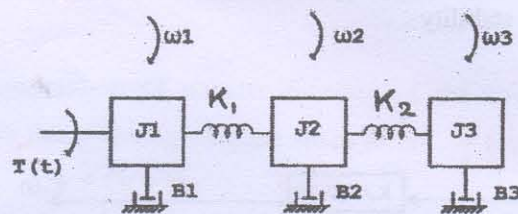


Figure (e)

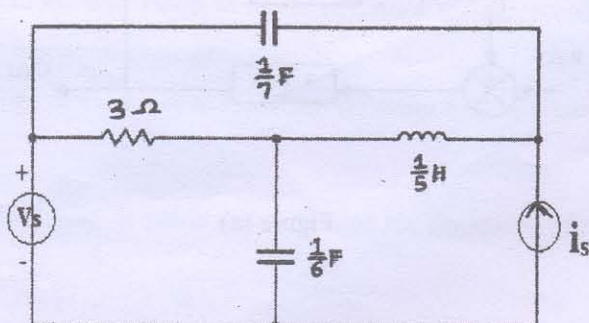


Figure (f)